Introducing the CodeQL Reference

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Introducing the CodeQL Reference: An Offline, AI-Ready Knowledge Base

Download the companion PDF:
CodeQL-Reference-Blog-Post.pdf

The Must Learn CodeQL repository now contains a structured, offline-searchable CodeQL reference designed for both humans and AI assistants. This post explains what it is, how to download and use it, how to contribute, and how to plug it into an MCP tool, Grok Bot, RAG pipeline, or custom CodeQL assistant.

What It Is

The reference lives at `reference/` in the MustLearnCodeQL repository. It is a collection of Markdown files plus a machine-readable `index.json` that catalogs:

- **CLI commands** — `database create`, `database analyze`, `query run`, pack commands, resolve commands, and more
- **QL language constructs** — predicates, classes, modules, formulas, aggregates, recursion, data-flow patterns, and `qlpack.yml`
- **Standard library highlights** — key classes and predicates for JavaScript, Python, Java, C/C++, C#, Go, Ruby, Rust, and Swift
- **Example queries** — ready-to-adapt patterns such as string-literal finders, empty-function detectors, class/struct listings, parameter listings, and source-to-sink skeletons

Everything is plain Markdown so it is easy to read, search, and embed.

How to Download

Clone the repository:

```
git clone https://github.com/rod-trent/MustLearnCodeQL.git
cd MustLearnCodeQL
```

The full reference is in the `reference/` directory. You can also download a single file or the entire tree from the GitHub web UI.

How to Search Offline

Because the content is plain Markdown, any local search tool works:

- **ripgrep**

  ```
  rg -i "database create" reference/
  ```

- **VS Code** — open the `reference/` folder and use the built-in search
- **Any Markdown viewer or editor** — the files are self-contained

The `index.json` file gives a quick catalog of every entry (title, path, summary) so you can jump to the right document without scanning the whole tree.

How to Use It

As a human reference

Browse the folders:

- `reference/cli/` — command references with synopsis, key options, and examples
- `reference/ql-language/` — language concepts
- `reference/libraries/<language>/` — overview, data-flow notes, and example queries

Start with `reference/README.md` for the layout.

As a knowledge source for an AI assistant

The reference is deliberately structured for AI consumption.

RAG pipeline

Point your embedding pipeline at the `reference/` tree. Chunk by file or by heading. The short summaries in `index.json` make good retrieval metadata.

MCP resource or tool

Expose the Markdown files (or the whole directory) as an MCP resource. An agent can list files via `index.json` and then fetch the specific document it needs.

Grok Bot

A CodeQL Assistant Grok Bot has already been created that is instructed to prefer this reference when answering CodeQL questions. You can point additional bots at the same files.

Custom CodeQL assistant

Load `index.json` plus the Markdown files into whatever retrieval or context mechanism your assistant uses. Because the content is version-controlled, you can pin an assistant to a specific commit for reproducibility.

How to Contribute

Contributions are welcome. The easiest ways to help:

1. **Add or improve a CLI command page** under `reference/cli/`
2. **Expand a language overview or data-flow page** under `reference/libraries/<language>/`
3. **Add example queries** under `reference/libraries/<language>/examples/`
4. **Update `index.json`** so new files are discoverable

Keep entries concise, example-driven, and accurate to the official CodeQL documentation. Open a pull request against the main branch.

Next Steps

- Explore the reference: github.com/rod-trent/MustLearnCodeQL/tree/main/reference
- Try the CodeQL Assistant Grok Bot
- Contribute an example query or a missing CLI command

The reference is a living document. As CodeQL evolves, so will this knowledge base—especially with community contributions.

Must Learn CodeQL — Rod Trent
Companion repository: github.com/rod-trent/MustLearnCodeQL